

Fall 2024
BIO 322 Comparative Animal Physiology
CRN 20548

Department of Biology
University of New England

Lecture Days/Time/Room: Tuesday and Friday / 12:30-1:50pm / Marcil 217

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Office Hours: Tuesday 2-4pm and by appointment

Course Description from the Course Catalog:

322 - Comparative Animal Physiology

This course will take a comparative approach to the physiology of animals. Students are introduced to a variety of animal systems and homeostatic and adaptive mechanisms. Adaptive mechanisms from the molecular to the organismal level will be considered.

Further course description: Animal physiology is “*the study of how animals work*”. Comparative physiology is the process of comparing and contrasting the mechanisms and details of physiology among different organisms. These comparisons can be among individuals exposed to different environments, or of different developmental stages (size, age), or those with different genetic make-ups—such as different sexes, populations, or species. We will examine a range of species in the context of homeostasis and adaptation. In addition, this course addresses the Critical Thinking theme of the Core Curriculum.

Course’s Role in the Curriculum: This course fulfills the requirement of

- a) Marine Sciences: Physiology area
- b) Biology: Physiology and Cellular area
- c) Animal Behavior: Physiology requirement
- d) Any other major: elective

Credit Hours: 3
Prerequisite: MAR105 and MAR106 or BIO104 or BIO105 and BIO106
Minimum Grade: C-
Co-requisite: n/a

Required Text: Animal Physiology, 5th ed. 2022. R.W. Hill, D.J. Cavanaugh and M. Anderson. Sinauer; older editions work as well

Other good text books:

Animal Physiology, Mechanisms and Adaptations. 2002. Eckert. W.H. Freeman and Company
Color Atlas of Physiology. 2003. A. Despopoulos & S. Silbernagl. Thieme

Biochemical Adaptation, Mechanism and Process in Physiological Evolution, 2002. P.W. Hochachka & G. N. Somero. Oxford University Press
Animal Physiology, Adaptation and Environment, 1997. K. Schmidt-Nielsen. Cambridge University Press

Relevant peer-reviewed journals (most of those are available as e-journals in the UNE library): Journal of Comparative Physiology A and B, Journal of Experimental Biology, Biological Bulletin, Science, Nature, American Journal of Physiology, Comparative Biochemistry and Physiology, Physiological and Biochemical Zoology and many others.

Student Learning Outcomes:

At the end of the semester you will be able to:

- 1) describe how animals function in different types of environments and maintain homeostasis;
- 2) discuss underlying physiological principles that govern animal life;
- 3) explain the intimate relationship between structure and function;
- 4) read, interpret and evaluate scientific journal articles;
- 5) write scientific journal-style research reports;
- 6) proceed to higher level marine sciences or biology courses.

Student Assessment:

Your achievement of course objectives will be based on three formal scientific papers, three midterm exams and a comprehensive final exam. The scientific papers will count for nearly half of your course grade. Activities that require your being absent from an exam or to miss an assignment due date may be considered excusable AND the work still needs to be complete as SOON AS POSSIBLE. Those who know in advance that they have a conflict at the time of an exam or collection of an assignment should see me NO LATER THAN 48 HOURS BEFORE THE CONFLICT to arrange for taking the exam or submitting the assignment.

You are responsible for all of the material that is covered in lecture and exams will be written with this in mind; therefore, it is in your best interest to attend all classes, read all assigned materials, and watch all posted videos, powerpoints, etc..

Your final grade will be calculated as follows:

Midterm Exams	3 x 14% =	42%
Papers	3 x 10% =	30%
Discussions (Perusal/In class)	9 x 2% =	18%
Cumulative Final Exam	1 x 10% =	10%

Expected Workload: Federal regulation defines a credit hour as an amount of work represented in intended learning outcomes and verified by evidence of student achievement that is an institutionally established equivalence that reasonably approximates not less than

1. One hour of classroom or direct faculty instruction and **a minimum of two hours of out-of-class student work** each week for approximately fifteen weeks for one semester or trimester.
2. At least an equivalent amount of work as required in paragraph (1) of this definition for other academic activities as established by the institution, including laboratory work, internships, practicals, studio work, and other academic work leading to the award of credit hours.

UNE's Core Values Statement acknowledges "the benefits of a richly varied campus community arise not only from the diversity of its people but also from a sense of belonging" (see [Mission, Vision, and Values](#)). The classroom should provide a space for learning, dialogue, and action that promotes the full engagement of students from all backgrounds. Instructors and students will strive to create a respectful and supportive environment for collaboration, empathy, and the building of meaningful relationships.

To help achieve an inclusive learning environment:

- If your name (or associated pronouns) differs from the roster, please let me know.
- If you anticipate needing to miss or shift class responsibilities due to a religious observance, please let me know.
- If you experience any form of discrimination in the classroom or on campus, you are asked to contact the Director of the Office of Title IX and Civil Rights Compliance, Angela Shambarger, Title IX Coordinator, ashambarger@une.edu , 207-221-4554, or online at <https://www.une.edu/title-ix/reporting> .

UNE Academic Policies and College of Arts and Sciences Policies and Student Information

See <https://www.une.edu/catalog/2024-2025/undergraduate-catalog/academic-policy-and-regulations> for general university undergraduate academic policies. See <https://www.une.edu/cas/academic-policies> for College of Arts and Sciences academic policies and procedures.

GRADING SCHEMA

All courses in the College of Arts and Sciences use the common grading scheme to convert weighted percentages to their corresponding letter grades:

COLLEGE OF ARTS AND SCIENCES GRADING SCHEMA

A	93.0-100%
A-	90.0-92.9%
B+	87.0-89.9%
B	83.0-86.9%
B-	80.0-82.9%
C+	77.0-79.9%
C	73.0-76.9%
C-	70.0-72.9%
D	60.0-69.9%
F	<60%

ACADEMIC INTEGRITY

THE UNE STUDENT HANDBOOK STATES:

The University of New England (UNE) values academic integrity in all aspects of the educational experience. Any behavior that provides a student with an advantage or provides another student with another advantage or disadvantage may be considered academic misconduct. Academic misconduct in any form undermines academic integrity and devalues the original works and contributions of others. It is the responsibility of all members of the university community to actively uphold academic integrity. ([UNE Student Handbook, 2024, p. 18](#)).

Academic misconduct includes, but is not limited to:

a. Cheating:

- Copying from another individual's academic work, test, quiz, or other assignment.

- Receiving, providing, and/or seeking assistance/aid from another individual to complete academic work, test, quiz, or other assignment.
- The use of materials or devices during academic work, test, quiz, or other assignment which are not authorized.
- Possession or use of current or previous course materials without the instructor's permission.
- Obtaining, or coercing another person to obtain, an unadministered test, test key, homework solution or computer program/software.
- Substituting for another person, or permitting another person to substitute for oneself, to complete academic work.
- Uploading, downloading, or accessing complete or incomplete academic work, test, quiz, or other assignment without the prior approval of the instructor.
- Falsifying research data, laboratory reports, and/or other academic work offered for credit.
- Altering and/or destroying the work of another student.
- Failing to comply with instructions given by the person administering the academic work, test, quiz, or other assignment that results in academic misconduct not enumerated above.

b. Plagiarism/Self-Plagiarism:

- The representation of words, ideas, illustrations, structure, computer code, other expression, or media of another as one's own and/or failing to properly cite direct, paraphrased, or summarized materials.
- The submission of the same academic work more than once without the prior permission of the instructor and/or failure to correctly cite previous work written by the same student.

c. Collusion

- Any unauthorized collaboration or attempted collaboration with another individual to complete academic work, test, quiz, or other assignment that results in similarities in the work, including, but not limited, to providing unauthorized assistance to another student and/or allowing another student access to completed academic work.

d. Falsifying Academic Records

- Altering or assisting in the altering of any official record of the University and/or submitting false information.
- Omitting requested information that is required for, or related to, any official record of the University.

e. Misrepresenting Facts

- Providing false grades or falsifying other academic information.
- Providing false or misleading information in an effort to injure another student academically.
- Providing false or misleading information in an effort to receive credit for attendance or a postponement or an extension on academic work, test, quiz, other assignment.

f. Violation of Professional Standards

- Any act or attempted act that violates specific Professional Standards or a published Code of Ethics. NOTE: Students may be held accountable under this policy based on the applicable standards of their college or school of enrollment, declared major, degree program, and/or pre-professional program.

g. Unfair Academic Advantage

- Any other action or attempted action that may result in creating an unfair academic advantage for oneself or may result in creating an unfair academic advantage or disadvantage for another student.

The University procedures for reviewing and reporting charges of academic misconduct are found on page 114 of the [2024 UNE Student Handbook](#). The College of Arts and Sciences policy on reviewing alleged acts of academic misconduct can be found at <https://www.une.edu/cas/academic-policies>.

MIDTERM ACADEMIC PROGRESS REPORTS

The University of New England is committed to the academic success of its students. At the midterm of each semester, instructors will report the performance of each student as SATISFACTORY (S) or UNSATISFACTORY (U). Instructors will announce when these midterm academic progress reports will be available for viewing via U-online. This early alert system gives all students important information about progress in their courses. Students who receive an UNSATISFACTORY midterm report should take immediate action by speaking with their instructor to discuss suggestions for improvement such as utilizing the services of academic advising, the Student Academic Success Center, Counseling Services, and Residential Education.

STUDENT ACADEMIC SUCCESS CENTER (SASC)

The Student Academic Success Center offers a range of free services to support your academic achievement, including tutoring, writing support, digital project support, learning support, and many online resources. To see and schedule available appointments go to <https://une.tutortrac.com> or visit the SASC. To access our online resources, including links, guides, and video tutorials, visit <https://une1.sharepoint.com/sites/SASC>.

STUDENTS WITH DISABILITIES

The University of New England is committed to creating a learning environment that meets the needs of its diverse student body and will make reasonable accommodations for students with qualified disabilities. Any student eligible for and needing academic adjustments or accommodations because of a disability is encouraged to request accommodations through the UNE Student Access Center. Registration with the Student Access Center is required before accommodation requests can be granted.

Visit <https://www.une.edu/student-access-center> for more information.

Course Outline with Topics, Dates & Assignments/Readings (Subject to Change)
(Chapter numbers below refer to Hill et al., *Animal Physiology* 5th ed.)

Week	Tuesday	Friday	Friday IN class discussion OR exam	Reading	UNIT	PERUSALL Due	PAPER Due
1		30-Aug	Dissecting primary literature papers	Ch 1	Intro - Organisms and Environment	Brown, 2006 Comments due Wed, Sept 4, 11:59pm - Submit via Perusall	Fri, Sept 13, 11:59 pm - Paper 1 (Chs 7,8,10) Submit via Brightspace Turnitin
2	3-Sep	6-Sep	Brown, 2006	Ch 2 and 5			
3	10-Sep	13-Sep	Jost et al., 2012	Ch 7,8 and 10	Metabolism	Jost et al., Comments due Wed, Sept 11, 11:59pm - Submit via Perusall	
4	17-Sep	20-Sep	EXAM 1 (Chs 1,2,5,7,8,10)				
5	24-Sep	27-Sep	Young, 1938	Ch 12, 13, 14, 15, 16	Integrating systems	Young, 1938 Comments due Wed, Sept 25, 11:59pm - Submit via Perusall	Fri, Nov 1, 11:59pm - Paper 2 (Chs 12,13,14,15, 16,19,20) Submit via Brightspace Turnitin
6	1-Oct	4-Oct	Winter et al., 2005			Winter et al. Comments due Wed, Oct 2, 11:59pm - Submit via Perusall	
7	8-Oct	11-Oct No class				Frøbert, et al. Comments due Wed, Oct 16, 11:59pm - Submit via Perusall	
8	15-Oct	18-Oct	Frøbert, et al., 2022				
9	22-Oct	25-Oct	EXAM 2 (Chs 12,13,14,15,16)				
10	29-Oct	1-Nov	Rummel et al., 2023	Ch 19 and 20	Muscles and Movement	Rummel et al. Comments due Wed, Oct 30, 11:59pm - Submit via Perusall	
11	5-Nov	8-Nov	York et al., 2018	Ch 23, 24, 25	Respiratory Physiology	York et al. Comments due Wed, Nov 6, 11:59pm - Submit via Perusall	Fri, Nov 22, 11:59 pm - Paper 3 (Chs 23,24,25, 27,28,29) Submit via Brightspace Turnitin
12	12-Nov	15-Nov	Gartner et al., 2010			Gartner et al. Comments due Wed, Nov 13, 11:59pm - Submit via Perusall	
13	19-Nov	22-Nov	Davis et al., 2024	Ch 27, 28, 29	Salt, Water, and Excretion	Davis et al. Comments due Wed, Nov 20, 11:59pm - Submit via Perusall	
14	26-Nov	29-Nov No class					
15	3-Dec	6-Dec	EXAM 3 (Chs 19,20,23,24,25)				
16	Final Exam Week Dec 9-13		Final Exam will be held during the Registrar assigned final exam period				